Electric Current

The *electric current* I (e.g. in a wire) is the amount of charge passing a given point per unit time. Current is measured in coulombs per second, or amperes A = C/s in SI units. It may not be uniformly distributed through a wire—more on this in a bit.

As an example, consider water flowing in a pipe: 

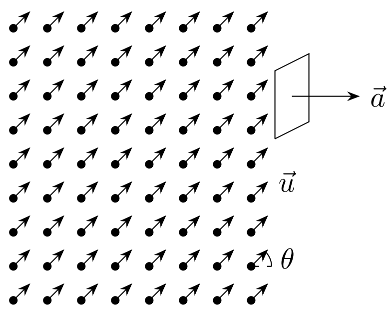
If the water is neutral, then the current is zero, but if it has some positive ions then the current would be positive (to the right). If it has negative ions, then the current would be negative (flowing to the left). By convention, current measures the flow of *positive charge*.

Current Density

First imagine there are a bunch of charges at some point in space, with charge q, number density n, and velocity $\vec{u}$. How many charges pass through a loop with area vector $\vec{a}$ in a time Δt?

Exercise 1: microscopic model of current density

(a) On the diagram, sketch the volume of space containing charges that *will* pass through the loop in some (short) time interval Δt.



(b) Calculate the volume of the region (in terms of $\vec{u}$, $\vec{a}$, and Δt), and the total charge enclosed.

$$V = \underbrace{\qquad\qquad}_{\text{cross-sectional area}} \times \underbrace{\qquad\qquad}_{\text{perpendicular height}}$$

$$= \qquad\qquad \text{in terms of vectors only}$$

(c) Calculate the current I passing through the loop.

$$I = \frac{Q_{\text{through}}}{\Delta t} =$$

(d) Generalize your result to the current through the loop resulting from multiple species k of charges q_k with number density n_k moving at $\vec{u}_k$.

$$I = \left(\qquad\qquad \right) \cdot \vec{a}$$

The final result motivates us to define the *current density* in terms of the physical motion of charges

$$\vec{J} = \sum_k q_k n_k \vec{u}_k$$

at each point in space, independent of any loop. Note that this can be a function of the position in space in general. The current flowing through any surface S (e.g. the cross-sectional area of a wire) is then

$$I = \int_S \vec{J} \cdot d\vec{a}$$

where the integral allows $\vec{J}$ to vary over the surface we are considering. This could even have been our definition of J and it's relation to the total current. The units of J are $[J] = \frac{\mathrm{C}}{\mathrm{s \cdot m^2}}$

Since this is linear in the velocity, if there are a bunch of charges with charge e all moving around with different velocities (the more realistic scenario) then,

$$J = ne\langle \vec{u} \rangle$$

where $\langle \vec{u} \rangle$ is their average velocity.

Charge Conservation

If charge is not building up anywhere (equilibrium) then the *total* current flowing through a *closed* surface must be zero: $\oint_S \vec{J} \cdot d\vec{a} = 0$.

Quick Exercise 2: Write this equilibrium condition in differential form.

More generally, if we allow charge to build up inside the surface, then we must have

$$\oint_S \vec{J} \cdot d\vec{a} = -\frac{d}{dt} Q_{\text{encl}}$$

by conservation of charge. (The sign comes because $d\vec{a}$ points outwards, but a net outward current from a region must *reduce* the charge enclosed.)

Quick Exercise 3: Write this more general condition in differential form.

This is the *local* statement of conservation of charge.

Conductivity and Ohm's Law

Now we turn to the most useful application: electric currents in materials. Recall that conductors have many free charges that can move. Surely, every material has at least a few free charges—maybe not enough to be considered a conductor, but enough that our discussion will be very general.

These free charges, as we already discussed, will be pushed by an electric field. Now we consider what happens *before* the charges reach static equilibrium (either during the short time before equilibrium, or because there is no static equilibrium—for example in an electric circuit).

It is seen *experimentally* that in most materials

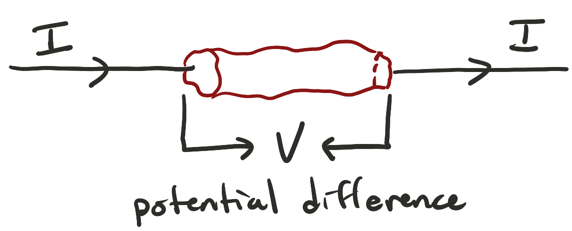
$$\vec{J} = \sigma \vec{E} \qquad \text{Ohm's Law}$$

where σ is a constant called "conductivity" that depends on the material. (Warning: this is NOT a surface charge density, sorry for the symbolic overload.)

Interesting properties of Ohm's Law:

- $\vec{J}$ is always proportional (as a vector!) to $\vec{E}$
- $\vec{J}$ is *linearly* proportional to $\vec{E}$, even though you might think $\vec{E}$ would cause *acceleration* of charges, increasing $\vec{J}$ with time.
- Rather than divide the world into conductors and insulators, it is more correct to say some materials have very high conductivity and some have very low (e.g. $\sigma_{\text{copper}} \simeq 10^8\ \text{C}^2/\text{N}\cdot\text{s}\cdot\text{m}^2$ and $\sigma_{\text{rubber}} \simeq 10^{-14}\ \text{C}^2/\text{N}\cdot\text{s}\cdot\text{m}^2$, where the units are also S/m i.e. Siemens/m, the SI unit of conductivity), as well as the full spectrum in between.
- In cgs units, $[\sigma] = \text{ s}^{-1}$; this is a hint that we should think of conductivity as a *timescale* over which the charges in a material move significantly. In fact, whether or not you call something a conductor depends on the timescale you are interested in.
- Ohm's Law will break down for high enough $\vec{E}$, or in specially designed materials (e.g. doped semiconductors).
- We also define the resistivity $\rho = \frac{1}{\sigma}$

Another way of writing Ohm's Law is more useful for circuit elements. For a chunk of material between two wires:



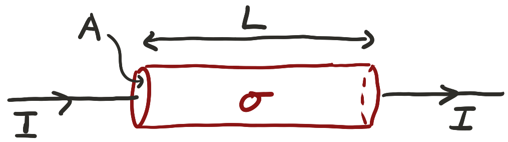
(Here we introduce V as the *voltage* across a circuit element; in electrostatics this is the same as the potential difference $\Delta\phi$.)

Since $V \propto E$ inside the material ($\times$ geometric factors) and $I \propto J$ ($\times$ geometric factors), we have

$$\vec{J} = \sigma \vec{E} \quad \implies \quad V = IR$$

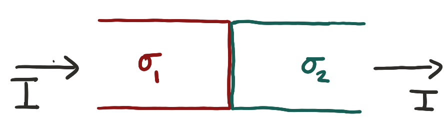
for some constant R that depends on σ and the geometry of the chunk of material.

Exercise 4: Find resistance R in terms of conductivity σ for a cylindrical chunk of material with length L and cross-sectional area A.



Exercise 5: Show that, if σ is constant and we wait for the current to reach (dynamical) equilibrium, the conductor is neutral everywhere inside its volume.

Exercise 6: Argue that if two wires with differing σ are connected, there will be net charge on the junction between the wires even after reaching equilibrium.



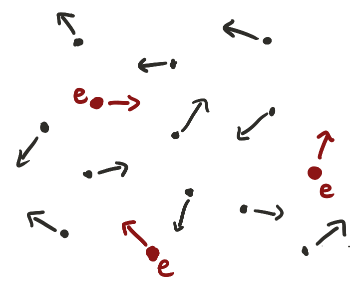
Physics of Electrical Conduction

Now we want to understand *why* many materials obey Ohm's Law, and how to relate σ to microscopic properties of the material. The problem we want to solve is: **if I apply an E-field across a material, how much curent will flow?**

To answer this we need to think more carefully about what is actually happening inside the material, especially with the free charge carriers. Recall most atoms in a material are neutral but a small fraction are ionized.

Let's start by considering a gas. The model will also apply to liquids and solids (more or less).

Consider the diagram to the right. There are many freely flying atoms all with large thermal velocities (~ 100 m/s at room temperature). A few are ions, with number density $n = \frac{\#\text{ ions}}{\text{m}^3}$, charge e, and mass M. Each atom (or molecule) flies a long distance with a constant velocity $\vec{u}$ before hitting another one in a hard collision and having its velocity completely randomized.



The average time between collisions is the *mean free time* which we'll call τ. (If it takes several collisions to randomize the velocity, just let τ be the total time required to randomize.) Now apply an electric field $\vec{E}$ across the gas. The ions each feel a force $e\vec{E}$, so each ion accelerates with $\vec{a} = \frac{e\vec{E}}{M}$.

Exercise 7: The Drude Model

(a) Consider a particular ion j, which has a velocity $\vec{u}_j^C$ just after its most recent collision, which happened a time t_j in the past. What is its present velocity?

$$\vec{u}_j = \vec{u}_j^C +$$

(b) Write an expression for the average velocity $\langle \vec{u} \rangle$ of many ions, indexed by j.

$$\langle \vec{u} \rangle = \qquad \text{as a sum}$$

$$= \qquad \text{using } \langle \cdots \rangle$$

(c) Using the assumptions that the $\vec{u}_j^C$ are random and uncorrelated, and $\langle t_j \rangle = \tau$, simplify your previous expression.[1] (Your result should only depend on e, $\vec{E}$, τ, and M—no more js.)

$$\langle \vec{u} \rangle =$$

(d) Derive expressions for the current density and conductivity in terms of the microscopic parameters (n, e, M, τ; the current density will also depend on the electric field $\vec{E}$).

$$\vec{J} = en \langle \vec{u} \rangle =$$

$$\sigma =$$

Interesting facts:

- We see that Ohm's Law works when E is small enough that the net drift velocity of the charges (caused by E itself) is much smaller than the thermal velocities.
- Note also that $\left\langle \vec{u}_j^C \right\rangle$ is not *exactly* zero and this is a source of noise (statistical uncertainty) in very sensitive electronics.
- Our crude model works pretty well. We build it for gas but it also mostly applies to solids and liquids.
  - e.g. glass or a crystal like NaCl are insulators. But when heated, their atomic structures become less rigid and ions can move some, increasing conductivity σ
  - in silicon (Si) and germanium (Ge), which are both semiconductors, raising the temperature liberates more electrons, increasing conductivity σ
  - In a metal, the atoms form a lattice and some electrons are very free to move, thanks to their wave-like nature (quantum mechanics).

    In Na there is one free conduction electron per atom: using our model and measured values we can conclude the mean free time is $\tau \simeq 3 \times 10^{-14}$ s; with an average thermal velocity of 10^5 m/s the mean free path is about 3×10^{-9} m, more than ten atoms!

    In fact, electrons wouldn't collide much at all except for (i) irregularities in lattice structure ('defects') which give (e.g.) copper a finite conductivity at $T = 0$ K; (ii) random thermal motion, which is why σ *decreases* with increasing temperature in metals.

    Below a certain temperature, some metals (e.g. lead, niobium) "superconduct:" they lose all resistance and become truly perfect conductors.

[1] Interestingly, $\langle t_j \rangle = \tau$ exactly, and not $\tau/2$ as you might expect since on average you are halfway between collisions. This is because the collisions happen randomly and independently without regard for how recently your previous collision happened.

Circuit Fundamentals

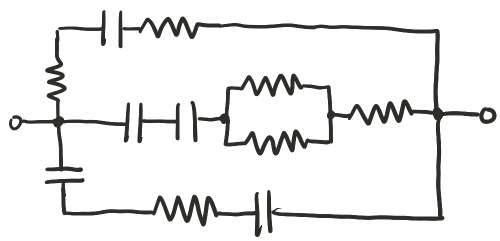
Very often, in the real world, we want to build a *circuit*: a closed loop of conductor that has various things in it (light bulbs, capacitors, etc.) and through which we will run current.

Each thing put into a circuit is called a *circuit element*. A surprisingly large number of such circuit elements can just be described as (i.e. they are electrically equivalent to) resistors or capacitors. For example, a light bulb is just a thin piece of material (the filament) through which we run current—in other words, a resistor.

To represent circuits schematically we use the following symbols:

- —— = wire, (assumed perfect conductor)
- = resistor, (finite σ, fixed geometry, R given)
- = capacitor, (fixed geometry, C given)
- $+$ $-$ = battery, (generates a given V (or $\Delta\phi$))

For example, this is a circuit:



We want to calculate how much current flows through each element, what the potential difference is across each element, how much charge is stored in capacitors, how much power is dissipated through resistors, and so on. Generally if we know all of the currents, we can figure out all the voltages (from Ohm's Law applied to resistors) and the rest will follow. But different strategies may be useful in different problems.

Kirchoff's Laws

You may already be familiar with Kirchoff's Laws for circuits:

- **Current/Junction Law:** The total current flowing into a junction of wires is equal to the total current flowing out from that junction.

- **Voltage/Loop Law:** The sum of all the voltages (potential differences) around a closed loop in a circuit is zero.

Strictly speaking, both of these laws hold only once the circuit has reached equilibrium, but we will often use them as long as currents are not changing rapidly. In Chapter 7 we will learn the limitations of these laws and how to properly generalize them.

Exercise 8: Prove the Current Law using conservation of charge in the form $\vec{\nabla} \cdot \vec{J} = 0$. *(Hint: Write the net current in/out of the junction as an integral.)*

Exercise 9: Prove the Voltage Law using what we have learned today, together with your knowledge of electrostatic fields.

Power Dissipation

Consider a charge q moving through a potential difference V. It takes $W = qV$ of work to (slowly) push that charge though the potential difference.

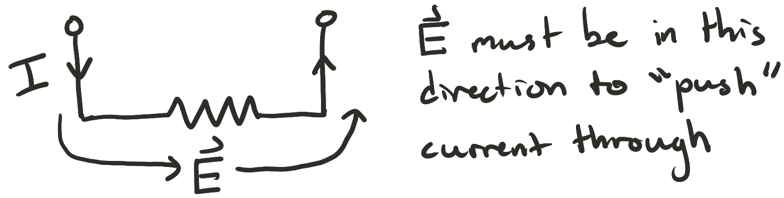
Exercise 10: Now consider many charges flowing in the wire, forming a current I. What is the *power* required to sustain that current, in terms of I and V?

Discuss: Interpret your result in the case of resistors, capacitors, and batteries. Where is the energy coming from or going to in each case?

Electromotive Force

Where does the energy in a circuit come from, if electric forces are conservative?

A useful way to think about this is: electric fields can move charge "downhill" in potential (as in a resistor), but by themselves they cannot keep pushing charge all the way around a closed loop forever. In steady operation, something must continually move charge back "uphill" in potential inside the source.



So inside a battery (or generator) there must be some additional *non-electrostatic* force on the charge carriers. At the circuit scale, we model this as an effectively *non-conservative* force that pumps charge from lower potential to higher potential.

- In a generator, this can come from mechanical input (e.g. rotation in a magnetic field).
- In a chemical battery, it comes from chemical reactions that separate charge.

These source forces are called *electromotive forces* (EMFs). Technically, the EMF is the *work done per unit charge* by the source mechanism as charge moves through the element. (So EMF is not itself a force.) Its units are Joules per Coulomb, i.e. Volts.

Exercise 11: Argue that, in a circuit element with an EMF, the magnitude of the EMF (represented by ε) is equal to the voltage (potential difference) across the element.

For this reason, the symbol ε is often used interchangeably with V for a battery: the EMF is specified by the voltage the source provides.

Activity 8: Circuit Applications